Ultra-Low Power, Ultra-High Sensitivity Omnipolar Hall Switch IC

1. Features

- Ultra-Low Power Omnipolar Hall Switch IC
- Typical average quiescent current at 3.3 V:
  - 1 Hz version: 0.1 μA
  - 20 Hz version: 1.8 μA
  - 500 Hz version: 35 μA
- Ultra-high sensitivity (typical B_{OP}): ±2.0 mT
- Wide voltage range: 1.8V to 5.5V
- Selectable push-pull and open-drain output
- Wide operating temperature range:
  - -40℃ to 105℃
- Package:
  - TO-92S (UA)
  - SOT23-3L (SO)

2. Applications

- Smart Home Appliances
- TWS earphone open/close detection
- Lid (or flip) open/close detection for phones and notebooks
- Valve position detection
- Liquid level detection
- Flowmeter

3. Description

SC2061/2 is an ultra-low power omnipolar Hall switch IC manufactured by CMOS process, featuring ultra-high sensitivity and excellent temperature stability.

SC2061/2 is designed for miniaturized, power-sensitive battery-powered systems and supports a wide operating voltage range of 1.8 V to 5.5 V. The device offers three sampling rates (1 Hz, 20 Hz, 500 Hz) to achieve ultra-low quiescent current, with selectable push-pull and open-drain output drivers for diverse applications. Its operating temperature ranges from -40 ℃ to 105 ℃, suitable for consumer electronics, smart home and industrial fields.

SC2061/2 is available in two packages: 3-pin through-hole TO-92S (UA) and 3-pin surface-mount SOT-23-3L (SO), with 100% lead-free matte tin-plated leads.



Fig.1 Package Outline

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4. Terminal Configuration

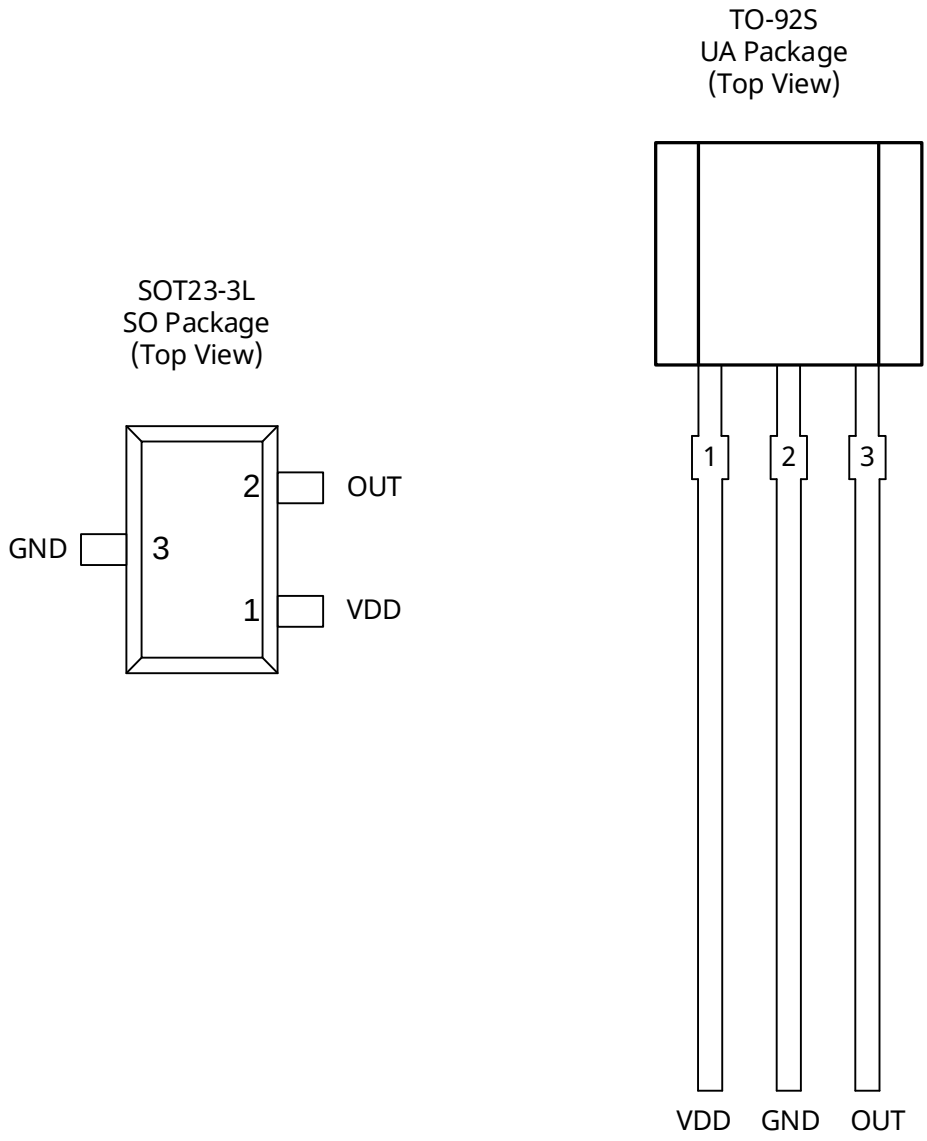


Fig.2 Pin Definition Diagram

Terminal			Type	Description
Name	UA	SO		
VDD	1	1	Power	1.8V to 5.5V power supply
GND	2	3	Ground	Ground terminal
OUT	3	2	Output	Push-Pull or Open-drain output. The open drain requires a pull-up resistor

5. Ordering Information

Ordering Information	Marking	Sampling Rate	Output	B_{OP}(mT)	B_{RP}(mT)	Package	Packing	Quantity
SC2061SO-LF-TR	2061	1 Hz	Push-Pull	±2.0	±1.0	SOT23-3L	Reel	3000/reel
SC2061SO-HF-TR	2061	500 Hz	Push-Pull	±2.0	±1.0	SOT23-3L	Reel	3000/reel
SC2061SO-TR	2061	20 Hz	Push-Pull	±2.0	±1.0	SOT23-3L	Reel	3000/reel
SC2061UA-BK	2061	20 Hz	Push-Pull	±2.0	±1.0	TO-92S	Bulk	1000/bag
SC2062SO-TR	2062	20 Hz	Open-Drain	±2.0	±1.0	SOT23-3L	Reel	3000/reel
SC2062UA-BK	2062	20 Hz	Open-Drain	±2.0	±1.0	TO-92S	Bulk	1000/bag

Ordering Information Format

SC206XSO-HF-TR

Packing
TR: Tape & Reel
BK: Bulk

Bandwidth Frequency
LF: 1Hz
Default: 20Hz
HF: 500Hz

Package Outline
SO: SOT23-3L
UA: TO-92S

Device Series
SC2061/2: Ultra-low power, high-sensitivity omnipolar Hall switch IC

6. Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted) (1)

Symbol	Parameter	Test Conditions	Min.	Max.	Units
V_{DD}	Power supply voltage		-0.3	6.0	V
V_{OUT}	Output terminal voltage		-0.3	6.0	V
I_{SINK}	Output terminal current sink		0	5	mA
T_A	Operating ambient temperature		-40	105	℃
T_J	Maximum junction temperature		-	105	℃
T_{STG}	Storage temperature		-55	150	℃

Note :
(1) Stresses above those listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods may affect device reliability

7. ESD Protection

Symbol	Parameter	Test Conditions	Min.	Max.	Units
V_{ESD_HBM}	HBM	According to: standard AEC-Q100-002 HBM	-2	+2	kV
V_{ESD_CDM}	CDM	According to: standard AEC-Q100-011 CDM	-750	+750	V

8. Thermal Characteristics

Symbol	Parameter	Test Conditions	Rating	Units
$R_{\Theta ja}$	UA Package thermal resistance	Single-layer PCB, with copper limited to solder pads	200(1)	℃/W
$R_{\Theta ja}$	SO Package thermal resistance	Single-layer PCB, with copper limited to solder pads	300(1)	℃/W

Note:
(1) Maximum voltage must be adjusted for power dissipation and junction temperature, see Thermal Characteristics

9. Operating Characteristics

9.1. Electrical Characteristics

over operating free-air temperature range V_{DD}=3.3V (unless otherwise noted) (1)

Symbol	Parameter	Test Conditions	Min.	Typ.(1)	Max.	Units
V_{DD}	Operating voltage (2)	$T_J < T_{J(Max.)}$	1.8	3.3	5.5	V
$I_{DD(Average)}$	Average Operating Current (LF)	V_{DD}=3.3V, T_A=25℃	-	0.1	0.3	μA
		V_{DD}=3.3V, T_A=105℃	-	0.2	0.5	μA
$I_{DD(Average)}$	Average Operating Current (Standard)	V_{DD}=3.3V, T_A=25℃	-	1.8	5.0	μA
		V_{DD}=3.3V, T_A=105℃	-	3.0	8.0	μA
$I_{DD(Average)}$	Average Operating Current (HF)	V_{DD}=3.3V, T_A=25℃	-	35	50	μA
		V_{DD}=3.3V, T_A=105℃	-	50	70	μA
t_{on}	Power on time	V_{DD}≥3.3V	-	30	55	μs
t_{ACTIVE}	Active time	V_{DD}=1.8V to 5.5V	-	100	200	μs
f_S	Magnetic Sampling Rate (LF)	V_{DD}=1.8V to 5.5V, T_A=25℃	-	1	-	Hz
f_S	Magnetic Sampling Rate (Standard)	V_{DD}=1.8V to 5.5V, T_A=25℃	-	20	-	Hz
f_S	Magnetic Sampling Rate (HF)	V_{DD}=1.8V to 5.5V, T_A=25℃	-	500	-	Hz
t_S	Magnetic Sampling Period (LF)	V_{DD}=1.8V to 5.5V, T_A=25℃	-	1000	-	ms
t_S	Magnetic Sampling Period (Standard)	V_{DD}=1.8V to 5.5V, T_A=25℃	30	50	100	ms
t_S	Magnetic Sampling Period (HF)	V_{DD}=1.8V to 5.5V, T_A=25℃	-	2	5	ms
t_r	Output rise time (10% to 90%)	R_L=1kΩ, Co=20pF	-	0.2	1.0	μs
t_f	Output fall time (90% to 10%)	R_L=1kΩ, Co=20pF	-	0.1	1.0	μs
Push Pull Output Characteristics						
V_{OH}	High-level Output Voltage	I_{OUT}=-1mA	V_{DD}-0.35V	V_{DD}-0.1V	-	V
V_{OL}	Low-level Output Voltage	I_{OUT}=1mA	-	0.1	0.3	V
Open Drain Output Characteristics						
I_{ILEAK}	Output Leakage Current	V_{DD}=5.5V, OUT=5.5V	-	-	0.1	μA
V_{OL}	Low-level Output Voltage	I_{OUT}=1mA	-	0.1	0.3	V

Note:

(1) Typical values are defined at T_A=25°C, V_{DD}=3.3V

(2) Maximum voltage must be adjusted for power dissipation and junction temperature, see Thermal Characteristics

9.2. Magnetic Characteristics

over operating free-air temperature range, V_{DD}=3.3V (unless otherwise noted)

Symbol	Parameter	Test Conditions	Min.	Typ.	Max.	Units
SC2061/2 ±2.0(1)/±1.0 mT(2)						
B_{OP}	Operated point	T_A=25℃	±1.0	±2.0	±3.5	mT
B_{RP}	Release point		±0.5	±1.0	±3.0	mT
B_{HYS}	Hysteresis		-	±1.0	-	mT

Note:
(1) Magnetic flux density, B is indicated as a negative value for North-polarity magnetic fields, and as a positive value for South-polarity magnetic fields
(2) 1mT=10Gs

10. Block Diagram

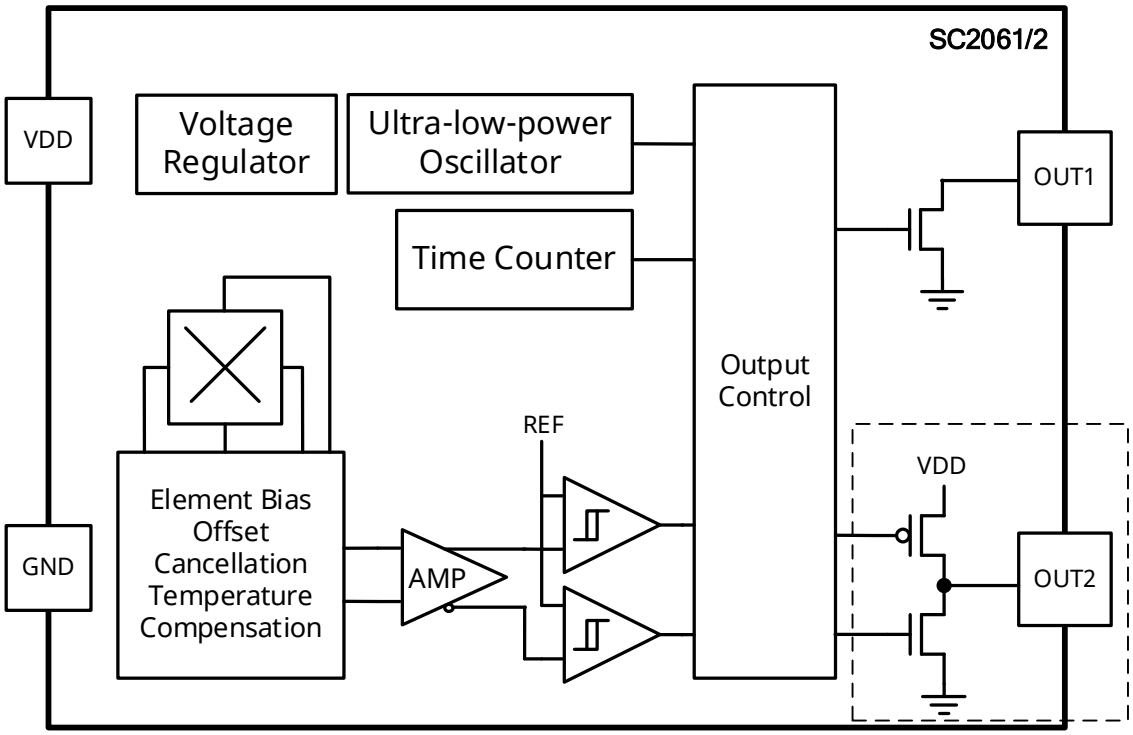


Fig. 3 Functional block diagram

11. Function Description

SC2061/2 is a low-power omnipolar Hall switch sensor designed for magnetic field detection. It responds to both south (S) and north (N) magnetic poles. Thanks to this feature, the device can detect magnetic fields of any polarity without distinguishing pole types. The IC integrates a Hall sensing element, analog signal conditioning circuitry and low-frequency oscillator. It operates over a supply voltage range of 1.8 V to 5.5 V and features ultra-low average power consumption. The chip samples magnetic flux density periodically and updates the output state, and enters low-power sleep mode between sampling intervals.

SC2061/2 is available in Push-Pull and Open-Drain output variants. For the Open-Drain version, an external pull-up resistor is required at the output pin, which can be pulled up to V_{DD} or an independent voltage domain. The output state is controlled by the magnetic signal: when the absolute magnitude of the magnetic field perpendicular to the Hall element exceeds the operate point (B_{OP}) threshold, SC2061/2 outputs a low level (ON); when the magnetic field decreases below the absolute value of the release point (B_{RP}), the device outputs a high level (OFF). The difference between the operate point and release point is defined as magnetic hysteresis (B_{HYS}). The built-in hysteresis effectively suppresses disturbances caused by external mechanical vibration and electrical noise.

11.1. Magnetic Field Direction Definition

A positive magnetic field is defined as the magnetic South pole facing the marked side of the package.

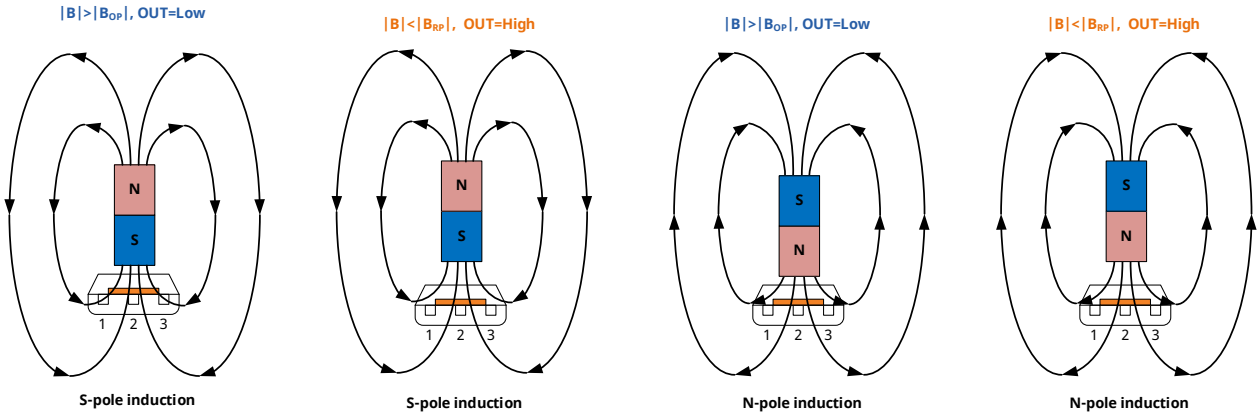


Fig.4 Magnetic Field Direction Definition

12.2. Transfer Function

SC2061/2 features omnipolar magnetic characteristics, meaning the device can be triggered by both south-pole and north-pole magnetic fields.

The output pulls low when the absolute magnetic field strength exceeds B_{OP}; the output releases when the absolute magnetic field strength falls below B_{RP}. Once triggered by a magnetic field, the device maintains a stable output state.

B_{OP}—magnetic threshold for activation of the device output, turning in ON (low) state

B_{RP}—magnetic threshold for release of the device output, turning in OFF (high) state.

$B_{HYS}= B_{OP}-B_{RP}$

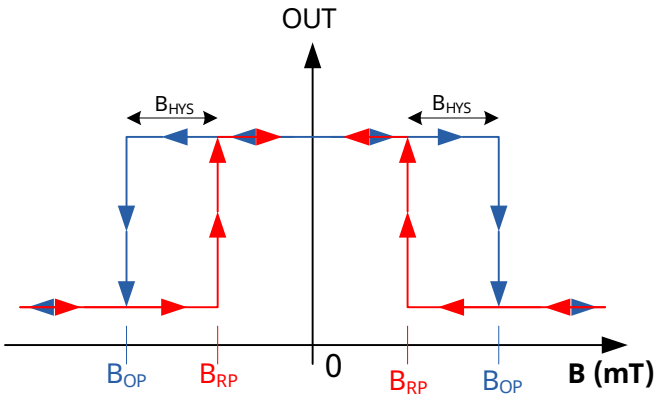
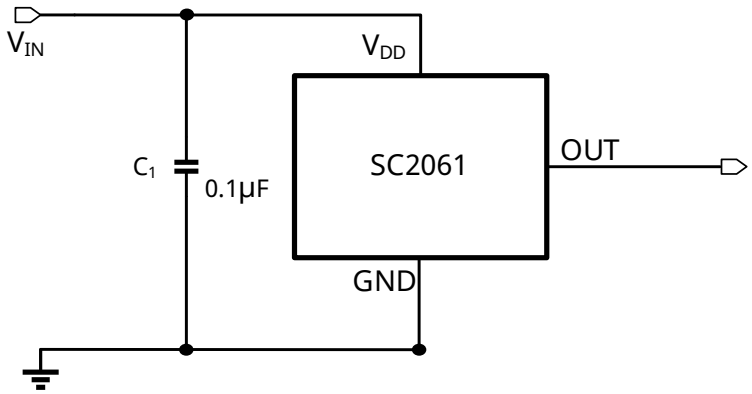


Fig.5 Transfer Characteristic Curve

12. Typical Application

Push-Pull Output



Open-Drain Output

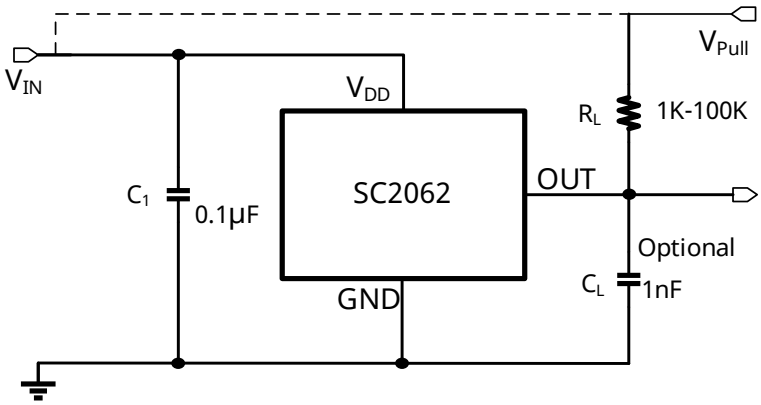


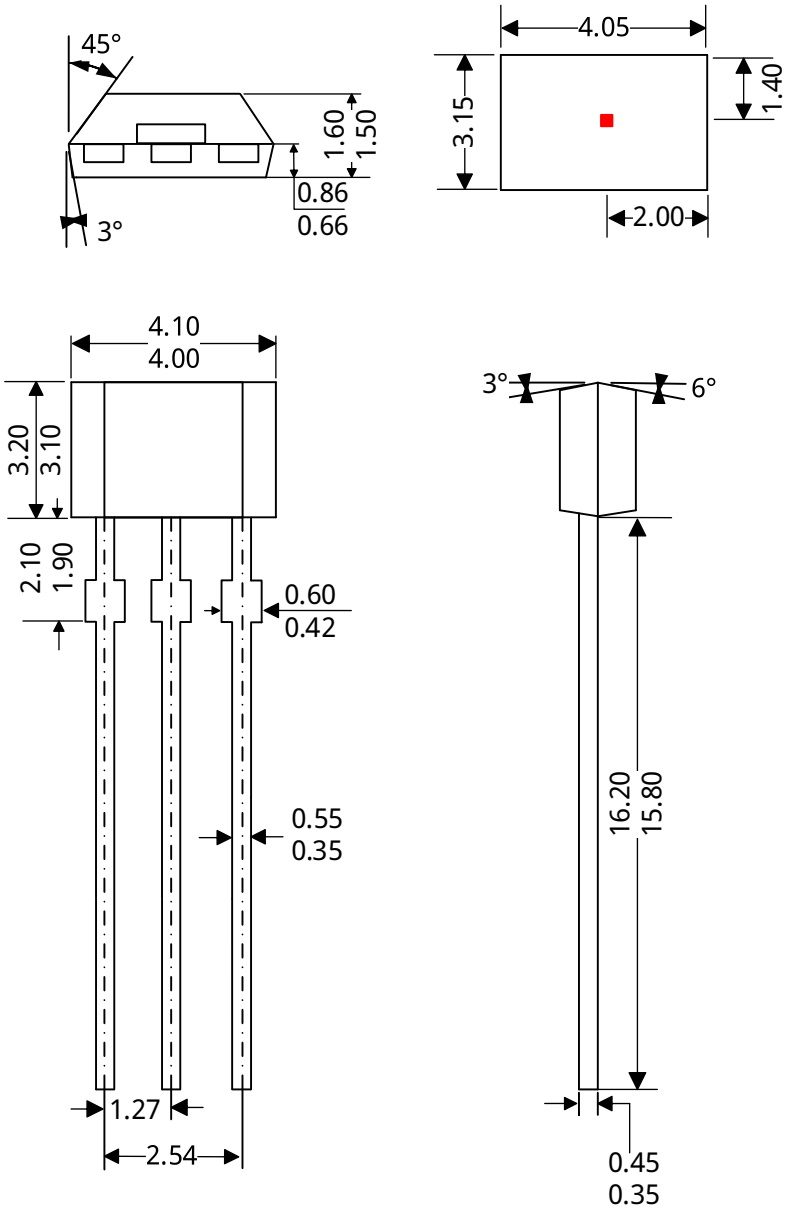
Fig.6 Typical Application Circuit

SC2061/2 integrates an internal voltage regulator and supports a wide operating voltage range. If the device is powered by an unregulated power supply, an external transient voltage protection circuit is mandatory. Even with a regulated power supply, it is recommended to add EMI/RFI suppression circuits to improve system noise immunity.

The chip utilizes chopper stabilization technology. To suppress external supply noise and interference introduced by internal chopping, a decoupling capacitor is strongly recommended to be placed close between the V_{DD}pin and GND. A 0.1 μF capacitor is suggested to be connected directly across the V_{DD} pin and ground to optimize filtering performance.

13. Package Information UA

TO-92S Package Outline Dimensions

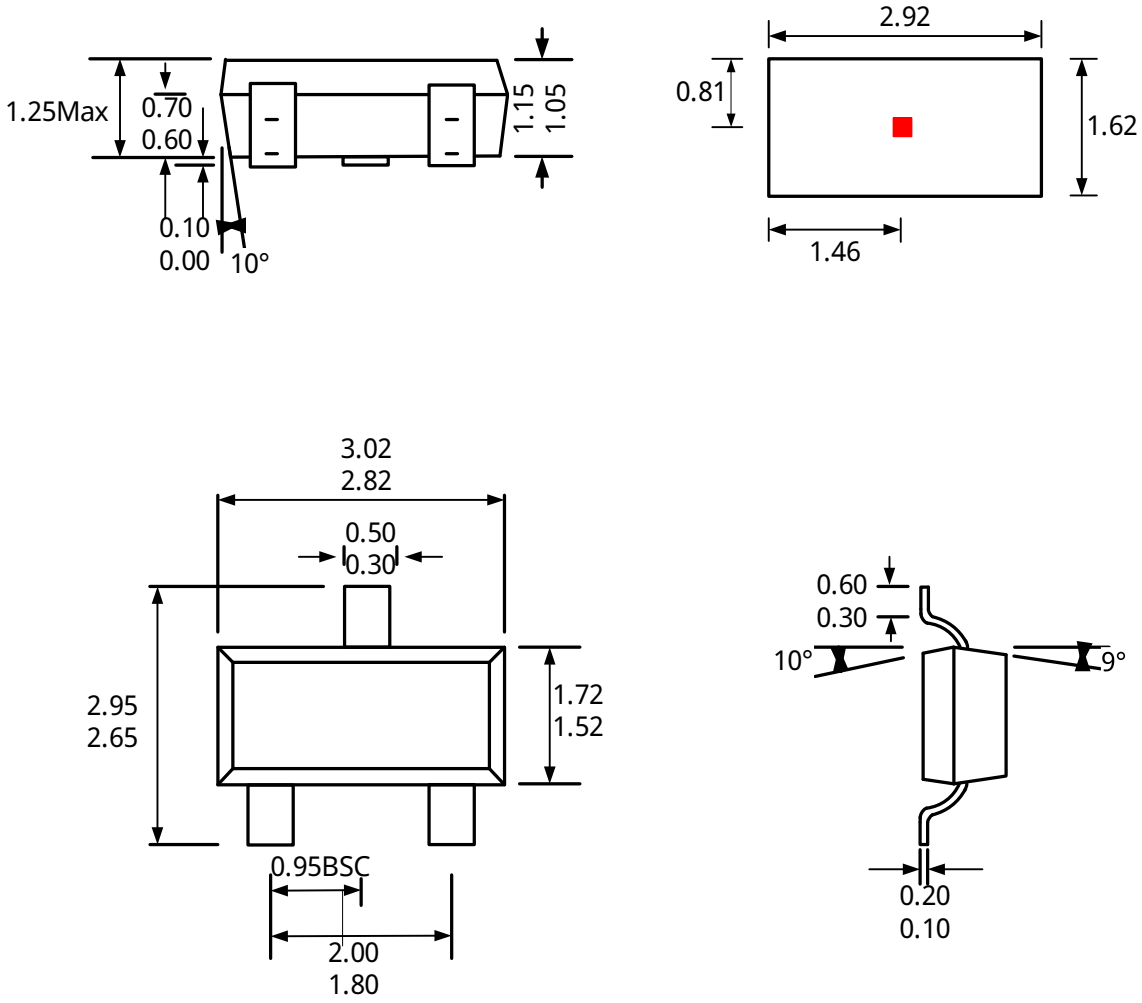


Notes:
(1)All unit in mm.
(2)Dimension does not include mold flash , protrusions or gate burrs.
(3)Allowable dambar protrusion shall be in excess at maximum material condition.

If no tolerance is specified , the dimension shall be theoretical reference value and shall not represent the exact dimension for actual measurement.

14. Package Information SO

SOT23-3L Package Outline Dimensions

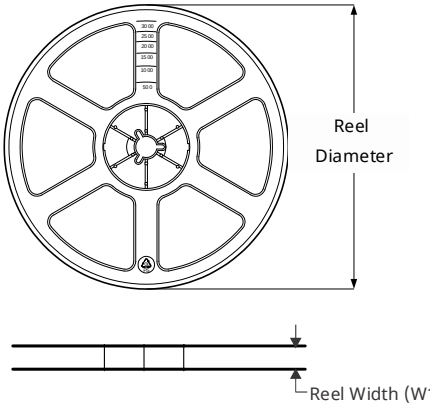


Notes:
(1)All unit in mm.
(2)Dimension does not include mold flash , protrusions or gate burrs.
(3)Allowable dambar protrusion shall be in excess at maximum material condition.

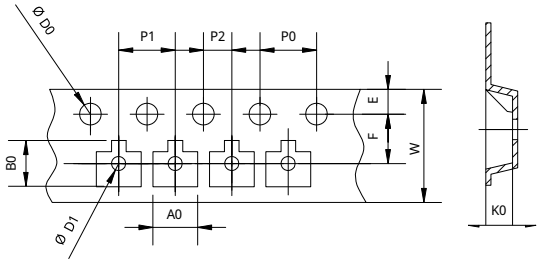
If no tolerance is specified , the dimension shall be theoretical reference value and shall not represent the exact dimension for actual measurement.

15. Tape&Reel Information

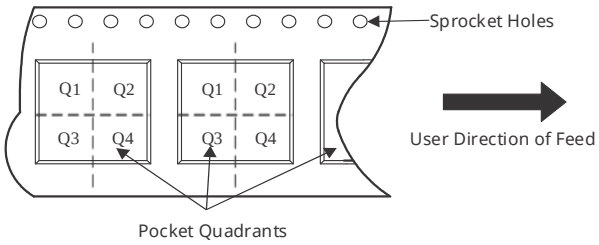
TAPE REEL DIMENSIONS



TAPE DIMENSIONS



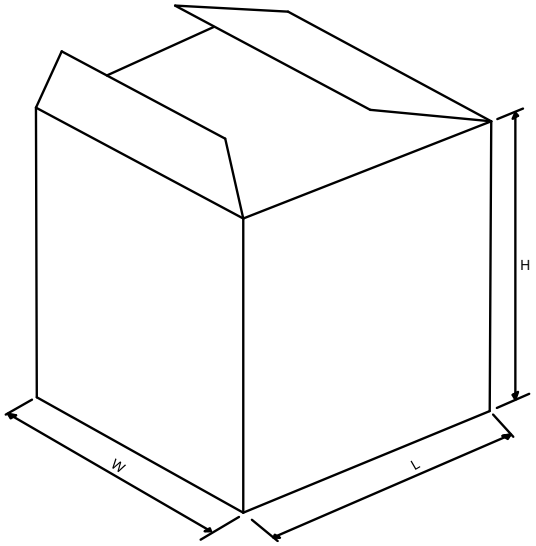
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Package Type	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	P0 (mm)	P1 (mm)	P2 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	W (mm)	Pin1 Quadrant
SOT23-3L	3000	180	8.4	4.00	4.00	2.00	3.18	3.28	1.32	8.00	Q3

16. Tape&Reel Box Dimensions



*All dimensions are nominal

Package Type	SPQ	Length (mm)	Width (mm)	Height (mm)
SOT23-3L	3000ea*10tape	210	210	210

17. Revision History

Revision	Date	Description
Rev0.1	2021-03-26	Preliminary datasheet
Rev.V1.0	2025-03-31	Official Version Released
Rev.V1.1	2026-08-11	Updated ordering table, supplemented packing information and document disclaimers

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